Lecture 3: Review, Applications

A non-singular example:

$$\begin{aligned} 1u + v + w &= ___ \\ 2u + 2v + 5w &= ___ \\ 4u + 6v + 8w &= ___ \end{aligned}$$

$$\downarrow \quad \begin{aligned} R_2 &\to R_2 - 2R_1 \\ R_3 &\to R_3 - 4R_1 \end{aligned}$$

$$\begin{aligned} u + v + w &= ___ \\ 3w &= ___ \\ 2v + 4w &= ___ \end{aligned}$$

$$\downarrow \quad R_2 \leftrightarrow R_3$$

$$\begin{aligned} 1u + v + w &= ___ \\ 2v + 4w &= ___ \\ 3w &= ___ \end{aligned}$$

System in <u>triangular form</u>
→ Always <u>exactly one</u> solution
→ # variables = # equations

An n by n matrix A is <u>nonsingular</u> if $Ax = b$ has one and only one solution $x \in \mathbb{R}^n$ for every $b \in \mathbb{R}^n$.

$$A = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 5 \\ 4 & 6 & 8 \end{bmatrix} \text{ is nonsingular.}$$

Elimination steps:
1) Replace a row by itself plus/minus a multiple of another.
2) Swap rows
3) Multiply a row by a nonzero number

$$\begin{aligned} x_3 + 4x_4 &= 6 \\ x_1 + 3x_2 + \quad + 2x_4 &= 1 \\ x_1 + 3x_2 + x_3 + 6x_4 &= 7 \end{aligned}$$

"Augmented matrix"

$$\left[\begin{array}{cccc|c} 0 & 0 & 1 & 4 & 6 \\ 1 & 3 & 0 & 2 & 1 \\ 1 & 3 & 1 & 6 & 7 \end{array}\right]$$

$$\downarrow \quad R_1 \leftrightarrow R_2$$

$$\left[\begin{array}{cccc|c} 1 & 3 & 0 & 2 & 1 \\ 0 & 0 & 1 & & 6 \\ 1 & 3 & 1 & & 7 \end{array}\right]$$

$$\downarrow \quad R_3 \to R_3 - R_1$$

$$\left[\begin{array}{cccc|c} 1 & 3 & 0 & 2 & 1 \\ 0 & 0 & 1 & 4 & 6 \\ 0 & 0 & 1 & 4 & 6 \end{array}\right]$$

$$\downarrow \quad R_3 \to R_3 - R_2$$

$$\left[\begin{array}{cccc|c} 1 & 3 & 0 & 2 & 1 \\ 0 & 0 & 1 & 4 & 6 \\ 0 & 0 & 0 & 0 & 0 \end{array}\right]$$

Echelon form

$$\left[\begin{array}{cccc|c} 1 & 3 & 0 & 2 & 1 \\ 0 & 0 & 1 & 4 & 6 \\ 0 & 0 & 0 & 0 & 0 \end{array}\right]$$

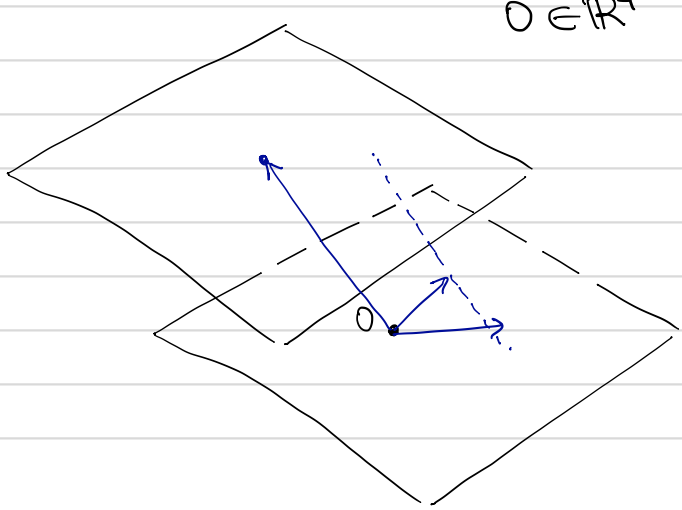
$$x_1 + 3x_2 + \qquad 2x_4 = 1$$
$$x_3 + 4x_4 = 6$$

Leading variables: x_1, x_3
Free variables: x_2, x_4

$$x_3 = 6 - 4x_4$$
$$x_1 = 1 - 3x_2 - 2x_4$$

All soln's: $\begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} 1 - 3s_1 - 2s_2 \\ s_1 \\ 6 - 4s_2 \\ s_2 \end{bmatrix}, \; s_1, s_2 \in \mathbb{R}$

$$= \underbrace{\begin{bmatrix} 1 \\ 0 \\ 6 \\ 0 \end{bmatrix}}_{\text{translate of}} + \underbrace{s_1 \begin{bmatrix} -3 \\ 1 \\ 0 \\ 0 \end{bmatrix} + s_2 \begin{bmatrix} -2 \\ 0 \\ -4 \\ 1 \end{bmatrix}}_{\text{Plane thru } 0 \in \mathbb{R}^4}$$



Theorem: A linear system can have either 0, 1 or ∞ many solutions.

Why: Do elimination:
- $0_c = 0 \rightsquigarrow 0$ sols.
- Free vars $\rightsquigarrow \infty$ sols
- Otherwise $\rightsquigarrow 1$ sol. ∎

Chemistry:

(caffeine)

$$x_1 \, C_8H_{10}N_4O_2 + x_2 \, O_2$$
$$\downarrow$$
$$x_3 \, CO_2 + x_4 \, H_2O + x_5 \, NO_2$$

Before = After

C: $8x_1 = x_3$
O: $2x_1 + 2x_2 = 2x_3 + x_4 + 2x_5$
N: $4x_1 = x_5$
H: $10x_1 = 2x_4$

$$\left[\begin{array}{ccccc|c} 8 & 0 & -1 & 0 & 0 & 0 \\ 2 & 2 & -2 & -1 & -2 & 0 \\ 4 & 0 & 0 & 0 & -1 & 0 \\ 10 & 0 & 0 & -2 & 0 & 0 \end{array}\right]$$

$\downarrow$ Elimination (p. 39)

$$\left[\begin{array}{ccccc|c} 2 & 2 & -2 & -1 & -2 & 0 \\ 0 & -4 & 4 & 2 & 3 & 0 \\ 0 & 0 & -1 & 0 & 2 & 0 \\ 0 & 0 & 0 & -2 & \frac{5}{2} & 0 \end{array}\right]$$

$$\left[\begin{array}{ccccc|c} 2 & 2 & -2 & -1 & -2 & 0 \\ 0 & -4 & 4 & 2 & 3 & 0 \\ 0 & 0 & -1 & 0 & 2 & 0 \\ 0 & 0 & 0 & -2 & \frac{5}{2} & 0 \end{array}\right]$$

$$\begin{aligned} 2x_1 + 2x_2 - 2x_3 - x_4 - 2x_5 &= 0 \\ -4x_2 + 4x_3 + 2x_4 + 3x_5 &= 0 \\ -x_3 \qquad\quad + 2x_5 &= 0 \\ -2x_4 + \tfrac{5}{2}x_5 &= 0 \end{aligned}$$

Free $\qquad$ Leading

$$x_5 = s \implies \begin{aligned} x_4 &= \tfrac{5}{4}s \\ x_3 &= 2s \\ x_2 &= \tfrac{27}{8}s \\ x_1 &= \tfrac{1}{4}s \end{aligned}$$

To get integers, set $s=1$:

$$2\,C_8H_{10}N_4O_2 + 27\,O_2 \longrightarrow$$
$$\longrightarrow 16\,CO_2 + 10\,H_2O + 8\,NO_2$$